# SynthoTranslate: Multimodal Knowledge Assimilation Tool for Overcoming Auditory Barriers and Bridging Language Barriers


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*Abstract*— **This paper introduces an automated system to improve educational accessibility through multilingual video translation and dubbing. It uses advanced speech-to-text (STT) technologies, natural language processing (NLP), and voice conversion models for accurate subtitles, translations, and dubbed audio in Indian languages. Key components include Whisper and Wav2vec for subtitle generation, GoogleTrans for translation, and Silero models for natural-sounding voice cloning. Indian Sign Language (ISL) transcription is integrated to support hearing-impaired students, with Wav2Lip ensuring precise lip-syncing of dubbed audio. Evaluation shows a Word Error Rate (WER) of 1.9% and cosine similarities over 95%. The system also offers content summarization and Q&A to enhance engagement. Future work will focus on optimizing speech recognition and improving ISL transcription for greater accessibility.**



***Keywords— Multilingual Translation, Voice Cloning, Indian Sign Language (ISL), Video Dubbing and Subtitling.***


## I. INTRODUCTION

Discovery Digital Network saw a 7.32% view increase by adding subtitles, emphasizing their value in educational content and reviews. Many Facebook videos are watched without sound, and one-third of internet users watch YouTube daily, with subtitles increasing engagement. Closed captioning boosts user time by 40%, and 80% of users prefer subtitles, aiding accessibility for the 360 million with hearing loss. However, challenges like speaker identification and timing persist. Voice dubbing retains emotional tone, crucial in education, but systems like DeepSpeech face memory and accuracy issues in longer videos.

Our project integrates customizable components into a single, free platform for diverse educational needs. Using RVC-based voice cloning, it outperforms paid alternatives, offering automatic summarization and interactive Q&A. Additionally, Indian Sign Language (ISL) is incorporated alongside subtitles, enhancing accessibility for hearing-impaired students. AI-driven algorithms personalize learning experiences, setting our approach apart.

India's linguistic diversity highlights the vital role of translation in connecting languages and preserving cultural heritage. While English is widely used, translation is essential for intercultural communication, especially online, and promotes social harmony. The NLP translation market is projected to reach US$5.92 billion by 2024, growing at an annual rate of 12.40%, and hitting US$11.94 billion by 2030. The U.S. will dominate with US$2,122 million by 2024[4]. These stats emphasize translation's importance in global connectivity. Previous systems often lacked accuracy and contextual depth, particularly for Indian languages [5]. The talks about the projection of the Multilingual Language Translation market by 2030.This also states the dire need of translation of different languages as a way of providing inclusive educational platforms to all.

### *A. Subtitle Summarization and Question-Answer Generation*

Subtitles help viewers quickly grasp the essence of video content, making a summary strategy vital for saving time. With data projected to exceed 180 zettabytes by 2025, concise summaries are essential for condensing vast information into digestible formats. In education, questions and answers boost engagement, comprehension, and critical thinking, offering valuable insights for instructors. However, challenges like accurately identifying verbs and matching tokens can affect answer precision. Limited answer extraction domains may hinder broader applicability, especially when using specific datasets like medical data, which can lead to inaccuracies in complex sentences. Despite achieving 80% accuracy, further improvements are needed to handle nuanced questions requiring deeper semantic understanding. [6].

### *B. Indian Sign Language(ISL) Translation*

The Americans with Disabilities Act (2016) mandates auxiliary aids like closed captions and subtitles for video accessibility, ensuring fundamental rights. The introduction of a universal knowledge assimilation tool is driven by the rapidly expanding education industry, with the global e-learning market expected to reach around


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$400 billion by 2026, doubling its size from approximately $200 billion in 2019. Notably, the learning management system (LMS) market alone generated over $18 billion in revenue in 2019.[9]

### *C. Voice Dubbing and Voice Clonning*

Voice dubbing plays a crucial role in education, especially in India's multilingual environment. Providing content in regional languages improves student accessibility and comprehension. Technologies like Retrieval-based Voice Conversion (RVC) replicate educators' voices, preserving their intonations and emotions for authenticity, while silero models for Text-to-Speech (TTS) synthesis ensure natural, contextually accurate audio. These solutions offer scalable, high-quality dubbing, allowing educators to reach wider audiences without compromising quality. This research addresses past challenges in domain-specific solutions and language accessibility, particularly for hearing-impaired individuals, by developing an interactive platform that supports personalized, multilingual learning. Using NLP and deep learning (DL), the platform enhances concept retention and breaks language barriers, making education more inclusive.

## II. Related Works

The paper proposes an automatic subtitle generation and semantic video summarization technique utilizing speech recognition and NLP-based text summarization algorithms. It introduces five summarization methods—Luhn's, LSA, Text Rank, Lex Rank, and Edmundson's algorithms—and demonstrates their effectiveness through experimental results, enhancing video navigation and accessibility on platforms like YouTube and Dailymotion in the era of big data [10]. The goal was to develop an LDA-based subtitle summarization model for educational videos, employing dataset expansion through human summaries, followed by LDA for extractive summarization. Performance comparisons showed LDA's superiority in ROUGE scores and human evaluation over TF-IDF and LSA models. Future work may address challenges with topics rich in numbers and verbs, focusing on length control and punctuation improvements [11]. The study examines subtitle segmentation parameters like BLEUbr and a new metric, Sigma, highlighting sensitivities to classification noise and estimating system results. Limitations include challenges in distinguishing homogeneous production and biases from alignment algorithms. Future work will involve developing balanced metrics and validating human decision links for improved subtitle segmentation assessment [12]. This research introduces a Python-based approach for efficient YouTube video summarization using the YouTube Transcript API and BERT algorithms, including extractive summarization and language translation, benefiting teaching, patent research, and information extraction [13]. It compares AI-based translation approaches with traditional machine translation, showing that AI translation is faster, more affordable, reliable, and accurate, highlighting its potential to enhance classroom learning [14]. An Android-based API powered by Firebase provides a solution to language barriers, enabling users to translate scanned text images offline using OCR and ML Kit technologies, improving user engagement [15]. The study aimed to generate computer-based exam questions aligned with learning goals using NLP tools like SpaCy and NLTK, with future plans for automating answer validation [16]. Another method used NLP to extract features like POS tags and synonyms, leveraging an Apache Solar server for Named Entity Recognition and question formation. This system was beneficial for single-word answers but faced limitations with similarly named proper nouns [17]. The research involved creating a Q&A system for a cohesive learning environment by extracting semantic relations and calculating word similarities, though it struggled with multi-part questions [18]. The paper also explores sign language processing, addressing challenges in recognition, translation, and modeling through interdisciplinary collaboration [19]. Finally, it presents EDUZONE, an educational tool that enhances online learning via video summarization and digital human technology, facing challenges with latency and accuracy [20].

## III. Proposed System

The system improves educational accessibility by automating video translation and dubbing in various Indian languages. It integrates STT, NLP, and voice cloning to generate subtitles and audio. The process includes audio extraction, transcription, translation, and subtitle synchronization, while voice cloning ensures natural-sounding audio. Other features include content summarization, question generation, and ISL transcription for hearing-impaired students.

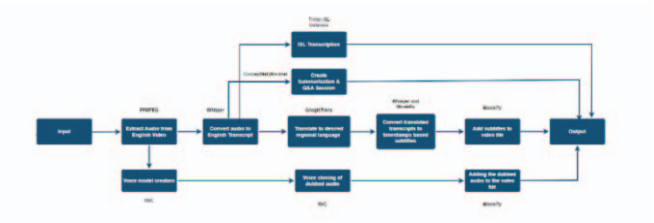


Fig 1: Flowchart of Proposed System

### *A. Subtitle Generation and Translation*

The subtitle generation starts with audio extraction via FFMPEG. Whisper and Wav2vec models convert speech to text, with Whisper excelling in noisy conditions. GoogleTrans handles translation into Indian languages using Google's neural machine translation. Whisper and MoviePy then sync the translated text with video timestamps, producing coherent subtitles for better comprehension.

### *B. Video Summarization and Question-Answer Generation*

Before Text Rank, BERT, and T5 integration, conventional summarization methods, reliant on word frequency and rule-based approaches, struggled with contextual comprehension and semantic nuances, often yielding incoherent summaries. Thus, a Question and Answering system was developed to clear any specific doubts that the user has in the video. This feature was implemented keeping in mind the need of QA systems for educational videos, where the student can use this feature to clear his/her doubts or might revise thus providing him/her a one-stop solution to test the concepts.

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*C. ISL(Indian Sign Language)Translation*

With over 70 million deaf individuals and 200+ sign languages worldwide, communication methods are crucial. Our prototype focuses on helping individuals with hearing disabilities, their families, professionals, and content creators. It starts by accepting .mp4 files or YouTube links, using Pytube to download and FFMPEG to extract audio as .mp3. The Whisper library, trained on 680,000 hours of data, acts as the ASR system for multilingual transcription and English translation. Whisper's encoder-decoder transformer architecture is illustrated in Fig 2.

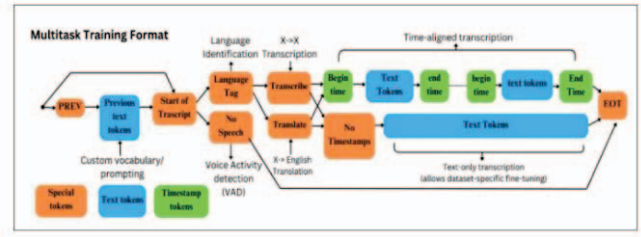



Fig 2: The encoder-decoder transformer of Whisper (a STT) model

The Whisper ASR system creates subtitles with accurate timestamps, while MoviePy handles burned-in subtitles. GoogleTrans, chosen for its language support, translates English into Indic languages using NMT models:

$$y^{\wedge} = \arg max_{y} P(Y|x) \qquad (1)$$

The y^ represents the translated text in the target language, with X as the input text and y as the possible translations, based on conditional probability. Subtitle summarization uses BART, T5, and TextRank algorithms, focusing on tokenization, encoding, and abstractive summarization. Post-processing ensures coherence. Fig 3 shows tokenization across these methods.

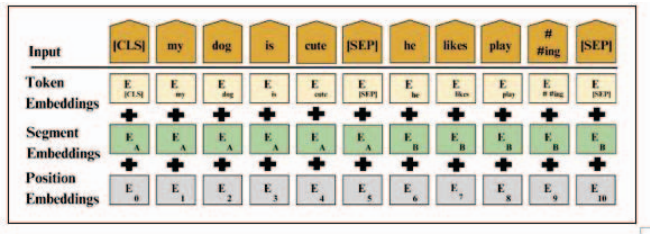



Fig 3: Tokenization of words in a sentence

The model is pre-trained on tokens "t," referencing past tokens "k" to compute the current token. To enhance tasks like summarization, it is fine-tuned in a supervised manner to maximize the probability of label "y" given feature vectors x1-xn. By combining these, an equation is formed where lambda (λ) regulates the effect of language modeling. In the summarization equations for BERT and T5, *X* represents the input, *Y* the summary, and *P* the conditional probability of generating the i-th token based on prior tokens. These equations maximize the conditional probability to generate a summary *Y* from text *X*.

$$L_3(C) = L_1(C) + L_3(C) \qquad (1)$$

TextRank tokenizes text, removes stop words, builds a similarity graph, and applies PageRank to select top sentences for extractive summaries. For MCQs, Sense2vec creates distractors using semantic word embeddings. WordNet's hypernyms provide abstract concepts (e.g., "India" → "countries in Asia"). Using "lava" as the keyword, we generated relevant distractors based on similar embeddings. ('molten_rock|NOUN',0.8014) and ('volcano|NOUN',0.7906). Methods like Sense2vec, WordNet, and ConceptNet initially gave incorrect options. Fine-tuning a BERT model on subtitles improved out-of-vocabulary handling using WordPiece tokenization, where each token $t_i$ is split into sub-word units, mathematically represented as: (Equation). The WordPiece tokenizer breaks tokens into sub-units S = ($S_1$, $S_2$, $S_n$), applying the tokenizer formula for each token:

$$S_i = \text{WordPTokenize}(t_i) = \{s_{i1}, s_{i2}, \ldots .s_{im}\} \qquad (4)$$

$$S_i = \text{WordPTokenize}(t_i) \ldots \text{WordPTokenize}(t_n) \qquad (5)$$

In BERT-based question answering, token scores for answer start and end positions are obtained with 'torch.argmax(start_scores)' and 'torch.argmax(end_scores)'. The identified span is sliced from the 'tokens' list and joined to form the answer.

*D. Voice Dubbing and Voice Clonning*

Voice dubbing is achieved using Silero models for text-to-speech (TTS) and Retrieval-based Voice Conversion (RVC) to replicate the original speaker's tone and style, ensuring natural and engaging audio in translated videos.

$$\text{PESQ Score} = 4.5 + 0.1 * (\text{Subjective Listening Quality}) \qquad (6)$$

These techniques ensure that the dubbed audio maintains the original context and emotional tone, enhancing the overall viewing experience.

- Inferencing Voice: Analyze voice characteristics using ML algorithms.
- Vocal Pitch Estimation: Isolate pitch of vocal components in music
- MDX-NET and De-echo: Reduce reverberation for clearer audio.
- Deep Invariant Optimal Quantization: Optimize audio quantization to reduce noise.
- Median Filtering: Minimize breathiness and noise for better quality.
- Transpose: Adjust audio pitch for correction or manipulation.
- Cloned Audio: Create modified audio that resembles the original.

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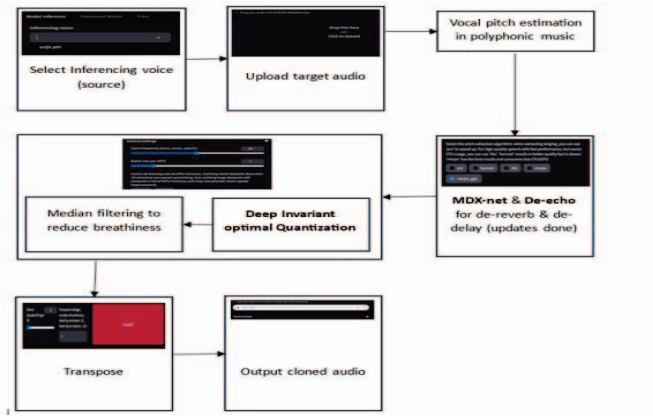



Fig 4: MDX-net, De-echo, DIOQ with median filtering in RVC

We fine-tuned ConceptNet, BART, and WordNet to enhance our Q&A system. ConceptNet provides commonsense knowledge for interpreting nuanced questions. BART generates accurate answers from Q&A datasets using its transformer architecture, while WordNet helps manage synonyms and disambiguate meanings. Together, these resources improve response precision and enrich the user learning experience.

### *A. Model Tunning*

In our research, we fine-tuned Retrieval-based Voice Conversion (RVC) models to improve their performance. RVC uses a vast voice database to convert source voices into target voices while preserving content. By fine-tuning with diverse datasets, we've enhanced the model's ability to replicate vocal traits like tone and pitch, ensuring high-quality voice cloning. We also integrated advanced algorithms for seamless voice adaptation, allowing real-time processing and natural speech synthesis. Optimizing hyperparameters and incorporating speaker-specific features enables personalized voice conversion, positioning our project as a robust tool for customizable voice outputs.

### *B. Lip-Syncing with Wav2Lip*

The system uses Wav2Lip technology with GANs to sync dubbed audio with lip movements, enhancing visual coherence in videos. This ensures authenticity in educational content, providing students with an engaging and seamless experience.

## IV. Results and Discussion

The following Table 1 represents a textual format of the preferred technologies used for the numerous features of the multimodal platform.

TABLE I.

| Task | Technology Experimented | Reason Used | Feature |
|---|---|---|---|
| Subtitle Generation | Whisper Wav2vec | Whisper-Works in noisy environments | Scrolling Subtitles |
| Translation | DeepL googletrans | Googletrans-100+ Languages DeepL-Only 35 Languages (None Indic) | Translation of Subtitles |
| Content Summarization | T5, BART and TextRank | BART for Context based summarization | Pipeline based content summarization |
| Question & Answering Model | T5, ConceptNet, WordNet and BERT | Context (Subtitle based Q-A) | MCQ Single blank answers |
| ISL-Indian Sign Language Translation | ML and NLP | Easy understanding for students with auditory impairments. | GIF based video of phrases/letters |

Table 1: Textual representation of the preferred technologies used for the system

The research achieved several goals, including the development of a deep-learning pipeline for accurate STT conversion to generate subtitles for videos. NLP algorithms were implemented for context generation, multilingual translations, and question-answering. Integration of ISL transcription enhances accessibility for users with hearing impairments.

### *A. Subtitle Generation*

The verification dataset contains .mp4 videos, 60 to 200 seconds long, mostly in English. It includes a variety of educational and entertainment content, ideal for subtitle generation and content analysis.

It presents examples of videos from the dataset used for subtitle generation. The speech-to-text output is synchronized with the video, including timestamps and keyword highlighting. Table 2 compares the Word Error Rate (WER) of different STT technologies. Due to the lack of domain-specific data, YouTube video examples were used to evaluate the fine-tuned model's performance

TABLE II.

| COMPARISON OF WORD ERROR RATE(WER) | | | | |
|---|---|---|---|---|
| **Video Title** | **Whisper (WER)** | | **Wave2Vec** | **Auto Generated by You tube** |
| | **Small** | **Medium** | | |
| Pre-Chewed Food Delivery \| Explainer Video | 8.51% | 5.32% | 38.54% | 30.93% |
| Animated Explainer Video Demo | 25.43% | 14.09 % | 36.86% | 23.08 % |
| Explainer Cafe's Animated | 1.9% | 0.6% | 55.13% | 34.62 % |
| PPC Adwords Animated Explainer Video | 4.05% | 0.00% | 42.07% | 25.09 % |
| Infographics Animation \| Explainer video production | 5% | 0.00% | 45.00% | 20% |
| How to hot-swap the smart heads and their positions Atlas Copco | 1% | 0.00% | 36.00% | 16.50 % |

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| | | | | |
|---|---|---|---|---|
| Atlas Copco - Home of Industrial Ideas | 7.64% | 0.69% | 48.61% | 22.92 % |

Table 2: Comparative Analysis of WERs for different methodologies

The Fig 5 shows the input video from the user and Fig 6 the output video respectively:

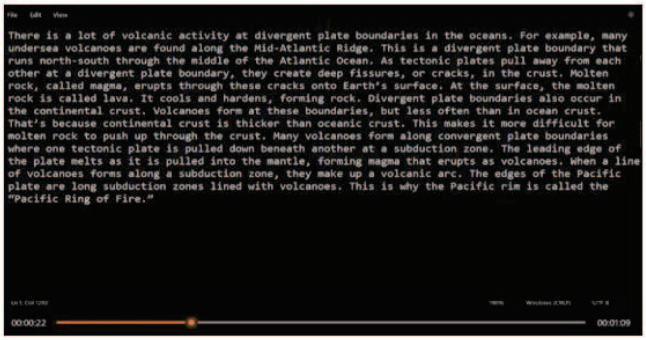



Fig 5: Input video (From the local device or YouTube Link)

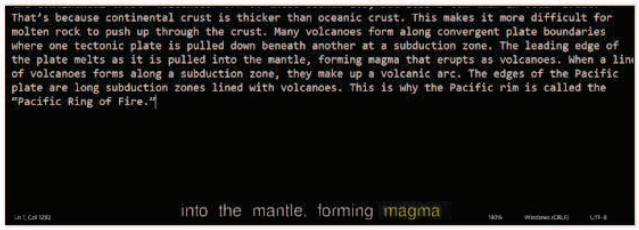



Fig 6: Output video

*B. Subtitle(Text) Summarization*

The outputs from three summarization methods—BART Extractive, T5, and TextRank—were analyzed. A GUI displayed results from BART, allowing users to summarize text beyond subtitles. Figure 7 shows the BART output, and Table 3 presents cosine similarity between original manual subtitles and those from our STT model.




Fig 7: Output-Box for BART based Summarization.

TABLE III.

| **Text Similarity of Actual Text to Generated text** | | |
|---|---|---|
| ***Cosine/Spacy Similarity*** | ***BART*** | ***T5*** |
| T5 | 96.59% | - |
| TextRank | 95.9% | 93.6% |
| BART | - | 88.78% |

Table 3: Cosine similarity of Original Subtitle to Generated Text

*C. Subtitle Translation*

The subtitle translation component uses Google Translate to convert English subtitles into various Indian languages, ensuring accuracy, contextual relevance, and clarity. Evaluation metrics showed strong performance: Translation Accuracy reached 92% by comparing automated translations with manual ones, reflecting high contextual correctness. Cosine Similarity revealed that automated translations exceeded 95% similarity to human translations in both context and content. In the equation, Ai and Bi represent the term frequency vectors of the original and translated texts, respectively.

*D. Question and Answer Generation*

Several methodologies were curated for creating a Question-and-Answer Generation model based on the subtitles namely MCQs and one-word type question and answers. Fig 8 and Fig 9 are MCQ based questions built on each and every line of the context(subtitles).

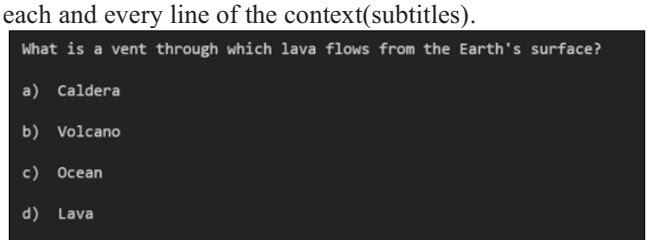



Fig 8: MCQ based on distractors used by ConceptNet

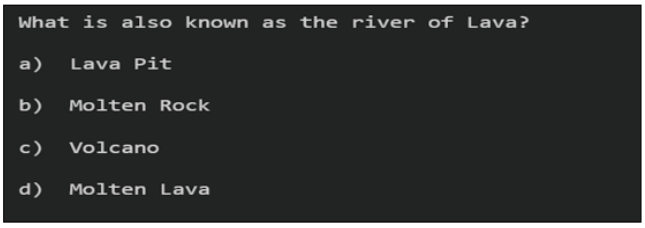



Fig 9: MCQ based on distractors used by Word-Net

*E. Indian Sign Language (ISL) Transcreption*

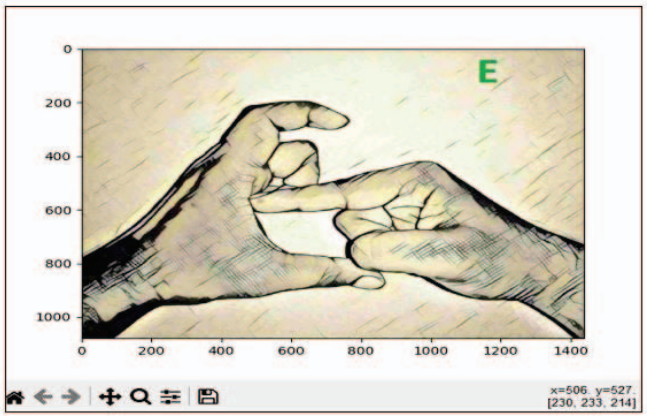



Fig 10: Letter based GIF representation using Tkinter

An essential feature is converting subtitles to sign language, catering to auditory impairments for a clear understanding at the user's pace. Fig 10 shows a GIF for words not in the dataset. Integrating Whisper for speech-to-text (STT) improved video effectiveness in noisy environments and enhanced multilingual support and real-time subtitles. However, optimizing speech recognition and video quality is crucial for longer videos. While there are challenges in providing subtitles in other Indic

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languages, the research has benefited students, especially those for whom English is a third language. Future efforts should focus on ISL translation and indexing GIFs from a government-approved database.

### *F. RVC Voice Clonning*

The F1F3syn measure quantifies breathiness in a voice signal. It begins by preparing the signal within a frequency range of 100-1500 Hz to capture relevant harmonics and normalizing the amplitude envelope for visual comparison. The measure is calculated by cross-correlating the amplitude envelopes of the first (F1) and third (F3) formants over a 25 ms segment. The synchronization index F1F3syn is derived, where higher values indicate modal voice quality and lower values suggest increased breathiness or creakiness.

$$Breathiness\ Factor = \alpha \cdot (F_3 - F_1) + \beta$$

In this model, **α** and **β** serve as coefficients that adjust the influence of **F1** and **F3** on the perceived breathiness of a sound. Here, **F1** refers to the first formant frequency, while **F3** represents the third formant frequency. The average F1F3syn value was approximately 0.911, indicating predominantly modal voice quality. This result highlights the effectiveness of F1F3syn in assessing breathiness and differentiating between modal and breathy/creaky voice qualities. Thus, F1F3syn serves as a valuable parameter for evaluating voice quality and understanding breathiness in vocal productions.

## V. Conclusion

This project successfully developed a multimodal and multilingual platform to make education accessible for all students, overcoming language barriers and supporting users with hearing impairments. However, limitations included restricted English language support for videos and ISL transcription for certain phrases. Future enhancements will focus on expanding subtitle support for more Indic languages, particularly Devanagari and Dravidian languages. Additionally, plans include implementing real-time human voice cloning for audio and developing a 3D model for ISL translation.

## References


[1] 80% of People Prefer Video Subtitles. Here's How they Affect Engagement. https://www.kapwing.com/resources/subtitle-statistics/.

[2] Akhil Kanade, Sourabh Gune, Shubham Dharamkar, Rohan Gokhale,"Automatic Subtitle Generation for Videos",International Journal of Engineering Research and General Science,vol. 3, Issue 6, 2015

[3] Prachi Sharma, Manasi Raj, Pooja Jangam, Sana Bhati, Prof. Neelam Phadnis, "Automatic Generation of Subtitle in Videos," SSRG International Journal of Computer Science and Engineering , vol. 6, no. 4, pp. 11-15, 2019, https://doi.org/10.14445/23488387/IJCSE-V6I4P103

[4] Language translation NLP - Worldwide, https://www.statista.com/outlook/tmo/artificial-intelligence/natural-language-processing/language-translation-nlp/worldwide.

[5] M. S. B. C., "Machine Translation Using Open NLP and Rules Based System 'English to Marathi Translator'", IJRITCC, vol. 2, no. 6, pp. 1730–1733, Jun. 2014.

[6] Shivani Singh,Nishtha Das,Rachel Michael,Dr. Poonam Tanwar,"The Question Answering System Using NLP and AI",International Journal of Scientific & Engineering Research vol. 7, Issue 12, December-2016.

[7] Priti Gumaste, Shreya Joshi, Srushtee Khadpekar, Shubhangi Mali,"AUTOMATED QUESTION GENERATOR SYSTEM USING NLP LIBRARIES",International Research Journal of Engineering and Technology (IRJET),vol. 07,Issue: 06,June 2020.

[8] Most popular video content type worldwide in 3rd quarter 2023, by weekly usage reach, https://www.statista.com/statistics/1254810/top-video-content-type-by-global-reach/.

[9] Size of the global e-learning market in 2019 and 2026, by segment(in billion U.S. dollars), https://www.statista.com/outlook/emo/online-education/worldwide.

[10] Aswin, VB & Javed, Dr. Mohammed & Parihar, Parag & Aswanth, K & Druval, CR & Dagar, Anpam & C V, Aravinda. "NLP Driven Ensemble Based Automatic Subtitle Generation and Semantic Video Summarization Technique",2021.

[11] S. S. Alrumiah and A. A. Al-Shargabi, "Educational videos subtitles' summarization using latent dirichlet allocation and length enhancement," Computers, Materials & Continua, vol. 70, no.3, pp. 6205–6221, 2022. https://doi.org/10.32604/cmc.2022.021780

[12] Alina Karakanta, François Buet, Mauro Cettolo, and François Yvon, 2 "Evaluating Subtitle Segmentation for End-to-end Generation Systems", In Proceedings of the Thirteenth Language Resources and Evaluation Conference, pages 3069–3078, Marseille, France. European Language Resources Association.,2022.

[13] Ilampiray P, Naveen Raju D, Thilagavathy A, Mohamed Tharik M, Madhan Kishore S, Nithin A.S, Infant Raj ,"Video Transcript Summarizer",E3S Web Conf. 399 04015,2023.

[14] Yu Yuxiu, "Application of translation technology based on AI in translation teaching," Systems and Soft Computing, vol. 6, pp. 200072, Jan. 2024, https://doi.org/10.1016/j.sasc.2024.200072.

[15] M. Vaishnavi, H. R. Dhanush Datta, V. Vemuri, and L. Jahnavi, "Language Translator Application," International Journal for Research in Applied Science & Engineering Technology (IJRASET), vol. 10, no. 7, pp. 45.98, Jul. 2022.

[16] Puneeth Thotad , Shanta Kallur , Sukanya Amminabhavi,"Automatic Question Generator Using Natural Language Processing", Journal of Pharmaceutical Negative Results, pp. 2759–2764, Dec. 2022.

[17] A. M. Agrawal, A. Atri, A. Chowdhury, R. Koneru, K. A. Batchu, and S. C. R. Mallavaram, "Question Answering System Using Natural Language Processing", IJRESM, vol. 4, no. 12, pp. 16–19, Dec. 2021.

[18] Maram Almotairi,Fethi Fkih,"Developing a Semantic Question Answering System for E-learning Environments using Linguistic Resources",Journal of Education and e-Learning Research,vol. 9, no. 4,pp. 224-232, 2022.

[19] Sarah S. Alrumiah, Amal A. Al-Shargabi,"Educational Videos Subtitles' Summarization Using Latent Dirichlet Allocation and Length Enhancement",Computers, Materials & Continua, vol. 70(3), pp. 6205-6221,2022,https://doi.org/10.32604/cmc.2022.021780.

[20] T. Wangchen, P. N. Tharindi, K. C. C. Chaveena De Silva, W. D. Thushan Sandeepa, N. Kodagoda and K. Suriyawansa, "EDUZONE – A Educational Video Summarizer and Digital Human Assistant for Effective Learning," 2022 7th International Conference on Information Technology Research (ICITR), Moratuwa, Sri Lanka, pp. 1-6,2022.


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